Mobile Phone Price Prediction using Machine Learning

Problem Statement

Predict the price range of mobile phones based on technical specifications using machine learning models.

Aim / Objective

To analyze mobile phone features and build machine learning models that accurately classify phones into different price ranges.

Dataset Information

The dataset contains mobile phone specifications such as RAM, battery power, clock speed, internal memory, camera details, and other hardware features.

Features and Target Variable

Features: RAM, battery power, clock speed, mobile weight, screen size, memory, camera, etc.
Target Variable: price_range

Technologies / Libraries Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost

Project Workflow

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Data Preprocessing
5. Model Training
6. Model Evaluation
7. Model Comparison
8. Final Prediction

Data Cleaning & Preprocessing

Checked missing values, duplicate records, feature scaling using StandardScaler, train-test split applied.

EDA Insights

- RAM strongly affects price range.
- Battery power also impacts pricing.
- Heatmap showed important feature correlations.
- Dataset was balanced across price categories.

Machine Learning Models Used

Logistic Regression, Decision Tree, Random Forest, XGBoost

Model Comparison

Logistic Regression: 97.5%
Decision Tree: 85%
Random Forest: 89%
XGBoost: 90.5%

Best Performing Model

Logistic Regression performed best with 97.5% accuracy because the dataset showed strong linear patterns between features and price ranges.

Evaluation Metrics

Accuracy Score: Measures correct predictions.
Confusion Matrix: Shows classification performance.
Precision, Recall, F1-score: Evaluate model quality in detail.

Important Visualizations

- Boxplots for feature distribution.
- Countplot for target class balance.
- Heatmap for feature correlation.
- Feature importance graph for Logistic Regression.

Final Conclusion

The project successfully predicted mobile phone price ranges using machine learning. Logistic Regression achieved the highest performance with strong prediction accuracy.

Challenges Faced

Understanding feature relationships, interpreting evaluation metrics, and selecting the best-performing model.

Future Improvements

Use larger datasets, apply hyperparameter tuning, try deep learning models, and deploy the project as a web application.

Key Skills Used

Data Cleaning, EDA, Feature Engineering, Machine Learning, Model Evaluation, Data Visualization

Model	Accuracy
Logistic Regression	97.5%
Decision Tree	85%
Random Forest	89%
XGBoost	90.5%